

Question

ReO_3F forms a disolvate **A** in anhydrous HF . **A** reacts with XeF_2 at -30°C to yield a 1:2 salt **B**, where $2/5$ of the XeF_2 is reduced during the reaction. **B** contains no hydrogen, with mass fractions of Xe and Re being 25.17% and 47.60%, respectively, and a molecular weight not exceeding 2000. The cation of **B** consists of only two elements and excludes fluorine, while the anion contains no Xe but features a fluorine-bridged structure, with the bridging fluorine atoms positioned trans to oxygen atoms. If the cation and anion are treated as free ions, all Re atoms exhibit identical chemical environments. **B** decomposes above -20°C to produce a Re-containing compound **C**. This reaction yields no ionic compounds, and no Xe–O bonds are present in the products. Throughout these processes, the oxidation state of the metal Re remains unchanged. In this context, if HF is generated as a reaction product, it is not considered a gas.

Which of the following statements is correct?

- A.** The relative molecular mass of **A** is 292.2
- B.** The reaction producing **B** and the reaction producing **C** have inconsistent gaseous products.
- C.** The chemical equation for the formation of **B** (with coefficients in the simplest ratio), where the sum of the product coefficients is an odd number.
- D.** The chemical equation for the formation of **C** (with coefficients in the simplest ratio), where the sum of the product coefficients is an odd number.
- E.** The cationic structure of **B** is trigonal.
- F.** The anionic structure of **B** exhibits fourfold rotational symmetry.
- G.** In the chemical equation for the formation of **C** (with coefficients in the simplest whole-number ratio), the coefficient of **C** is 4.

H. Among the elements that constitute **C**, there are two elements with the same atomic number.

Answer

Correct Answer: **G**

Detailed Explanation

ReO_3F forms a disolvate **A** in anhydrous HF , so the chemical formula of **A** is clearly $\text{ReO}_3\text{F}(\text{HF})_2$, with a relative molecular mass of 293.2, making option A incorrect.

CHECKPOINT

1 PTS

The chemical formula of **A** is $\text{ReO}_3\text{F}(\text{HF})_2$, with a relative molecular mass of 293.2

$\text{ReO}_3\text{F}(\text{HF})_2$ reacts with XeF_2 to form a 1:2 salt **B**, with 2/5 of the XeF_2 being reduced. The reduction product is clearly Xe .

CHECKPOINT

0.5 PTS

The reduction product is Xe

Assuming the reaction requires 5 molecules of XeF_2 , then **B** contains 3 Xe atoms. Based on the molecular weight not exceeding 2000 and the mass fraction information, it can be concluded that **B** indeed contains 3 Xe atoms.

CHECKPOINT

1 PTS

B contains 3 Xe atoms

According to the mass fraction information, the atomic ratio of Xe to Re in **B** is $(0.2517/131.3 : 0.4760/186.2) = 3 : 4$, so **B** contains 4 Re atoms.

CHECKPOINT

0.5 PTS

The atomic ratio of Xe to Re in **B** is 3 : 4

CHECKPOINT

1 PTS

B contains 4 Re atoms

Thus, the molecular weight of **B** can be calculated as $3 \times 131.2/0.2517 = 1565$.

CHECKPOINT

1 PTS

The molecular weight of **B** is 1565

B contains no hydrogen, and based on the reactants in the formation reaction of **B**, it can be inferred that it also contains oxygen and fluorine. The remaining molecular weight of **B** after accounting for Xe and Re is 426, which corresponds to exactly 10 oxygen atoms and 14 fluorine atoms. Therefore, the chemical formula of **B** is $\text{Xe}_3\text{Re}_4\text{O}_{10}\text{F}_{14}$.

CHECKPOINT

2 PTS

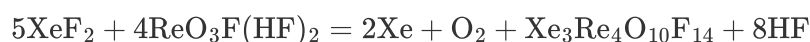
The chemical formula of **B** is $\text{Xe}_3\text{Re}_4\text{O}_{10}\text{F}_{14}$

In the reaction forming **B**, since the Re in **A** is already in its highest oxidation state, only the oxygen is oxidized by Xe, and the oxidation product is clearly oxygen. Thus, the equation for the formation of **B** can be written as:

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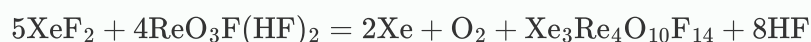
0.5 PTS

The oxidation product is clearly oxygen



CHECKPOINT

1 PTS



The sum of the coefficients of the products in the equation is 12, which is not odd, so option C is incorrect.

The anion of **B** does not contain Xe, so the cation contains 3 Xe atoms and no fluorine. Since **B** is a 1:2 ionic compound, the cation must carry a +2 charge. The 3 Xe atoms in the cation are not reduced and are in the +2 oxidation state, requiring an additional 4 negative charges, which can be balanced by two oxygen atoms. Therefore, the chemical formula of the cation in **B** is $\text{Xe}_3\text{O}_2^{2+}$, and its structure is bent, with oxygen atoms serving as bridging atoms. Thus, option E is incorrect.

CHECKPOINT

2 PTS

The chemical formula of the cation in **B** is $\text{Xe}_3\text{O}_2^{2+}$, and its structure is bent

Therefore, the anion of **B** carries a single negative charge and there are two of them, with the chemical formula $\text{Re}_2\text{O}_4\text{F}_7^-$.

CHECKPOINT

1 PTS

The chemical formula of the anion in **B** is $\text{Re}_2\text{O}_4\text{F}_7^-$

The anion of **B** contains fluorine bridges. Re in the +7 oxidation state commonly forms octahedral coordination with six ligands, so each Re is connected to two terminal oxygen atoms, three terminal fluorine atoms, and one bridging fluorine atom, forming two octahedra sharing a vertex.

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1 PTS

The structure of the anion in **B** consists of two octahedra sharing a vertex

However, it can be observed that each Re is bonded to two oxygen atoms and four fluorine atoms. Since one fluorine atom is bridging, the resulting octahedron cannot have four atoms coplanar and thus lacks fourfold rotational symmetry, making option F incorrect.

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1 PTS

The resulting octahedron cannot have four atoms coplanar

The decomposition of **B** yields a Re-containing compound **C**. The reaction products contain no ionic compounds, and no Xe – O bonds are present, so the Xe products are in the form of elemental Xe and XeF₂. Since the oxidation state of Re does not change, the reaction can be interpreted as Xe oxidizing oxygen to produce O₂, and the chemical formula of **C** can be written as **Double subscripts: use braces to clarify**, satisfying $2x + y = 7$. Given that Re in **B** is bonded to two oxygen atoms, assuming $x = 2$, the equation for the formation of **C** is:



CHECKPOINT

2 PTS



Thus, the chemical formula of **C** is ReO₂F₃, making option H incorrect.

CHECKPOINT

1 PTS

The chemical formula of **C** is ReO₂F₃

The sum of the coefficients of the products in the formation reaction of **C** is 8, which is not odd, so option D is incorrect. The gaseous products of this reaction are the same as those in the formation reaction of **B**, making option B incorrect. The coefficient of **C** is 4, so option G is correct.

In conclusion, option G is correct.

CHECKPOINT

1 PTS

Option G is correct